

- * In PMF we can have (2) or (3) heads but NOT 2.5 heads, so, no inbetween values are allowed.

Random variables can take only separate, countable values.

PROBABILITY MASS FUNCTION :- gives $P(X=x)$

- USED FOR DISCRETE RANDOM VARIABLES. or PMF tells us how likely each possible outcome is for a random event that has set values like rolling dice or flipping coins.
- It only works when there are clear, separate values the result can be (like 1, 2, 3, 4, 5, 6 for a die), not when any number in a range is the result.
- for example: When rolling a fair 6-sided die, the chance (probability) of getting a 'three' is 1 out of 6 or $(1/6)$.

- * Example :- When tossing four fair coins, we have the following possible outcomes for the number of heads (H) :-

- 0 head (TTTT)
- 1 head (HTTT, THTT, TTHT, TTTH)
- 2 heads (HHTT, THHT, TT HH, HTTH, THTH, HTHT)
- 3 heads (HHHT, HTHH, TTHH, HHTH)
- 4 heads (HHHH)

∴ Total count of sample space = 16

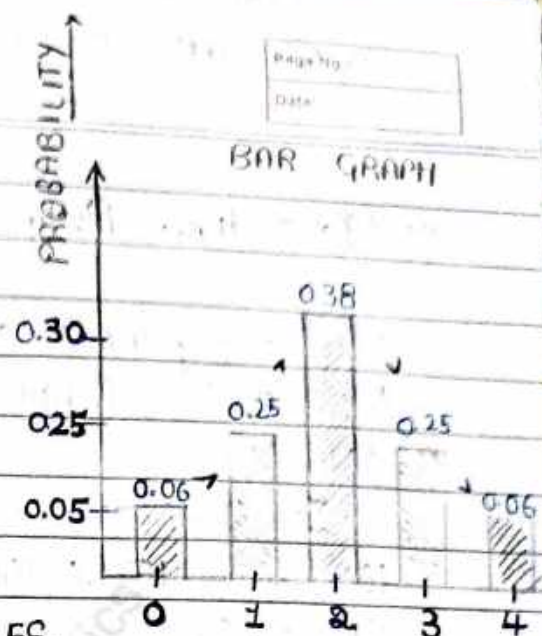
$$\therefore \text{PMF (0 Head)} = P(0) = \frac{1}{16} = 0.06$$

$$\text{PMF (1 Head)} = P(1) = \frac{4}{16} = 0.25$$

$$\text{PMF (2 Head)} = P(2) = \frac{6}{16} = 0.38$$

- $PMF(3 \text{ HEADS}) = P(3) = \left(\frac{4}{16}\right) = 0.25$

- $PMF(4 \text{ HEADS}) = P(4) = \left(\frac{1}{16}\right) = 0.06$



PROBABILITY DENSITY FUNCTION :-

- USED FOR CONTINUOUS RANDOM VARIABLES,
 ↳ It is a variable that can take any value in a range, not just fixed jumps. Example :- Height of a person can be 160.2 cm, 160.25 cm, 160.251 cm, etc (infinite possible values b/w 160 and 161 cm)

• PROBABILITY DENSITY FUNCTION (PDF) :-

This is a curve/function $f(x)$ that tells how likely the values around ' x ' are for a continuous random variable.

- We do not get the probability of an exact value like $P(X=2)$ from PDF, we get probability from an interval like $P(1.5 < X < 2.5) = X$

Q What is this curve/function $f(x)$?

A) For a continuous random variable, we draw a smooth curve on a graph; this graph is called $f(x)$, the probability density function curve.

On the horizontal axis (x-axis) we put all values of the variable (for example, all possible heights of people) and on the vertical axis (y-axis) we plot $f(x)$.

So इसका मतलब अगर x के किसी Range में curve की height ज्यादा है, तो उसी width के दूसरे Range के comparison में उस Range में x के गिरने की PROBABILITY ज्यादा होती है।

Q. "Tells how likely the values around 'x' are" - meaning.

- Pick any point 'x' on the x-axis, for example $x = 160$ cm height
- Look at the height of the graph/curve at that point

$f(160)$:

If the curve is high there it means value near 160 (like 159.8, 160.2, 160.5) are more common or more likely. If at some other point say $x = 190$ the curve $f(190)$ is very low, then values near 190 are much less likely compared to values near 160.

- So "how likely the values around x are" means :-
a higher curve at $x \Rightarrow$ the random variable usually falls near that x ; a lower curve at $x \Rightarrow$ it rarely falls near that x .

* IMPORTANT :- It is not probability of exact value

(For continuous variables, the probability at a single exact point $P(X=x) = 0$, $\therefore f(x)$ itself is not the probability.

To get real probability, we take a small interval around x , like from a to b and find the area under the curve between a and b :

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

$\therefore$ PDF = function, Probability = area under that function between 2 points a and b .

PDF hume batata hai ki, kaun-se 'x' ke aas pass values relatively more likely hain (curve Jahan High hai), aur kisi bhi range mein value aane ki real probability, us range ke neeche wale area (integral) se nikalti hai.

Q. Imagine a call centre that receives a large volume of customer inquiries and service requests. Customers are placed on hold until they can be connected to a customer service representative. The time a customer spends on hold is referred to as 'WAIT TIME'.

The mean wait time (μ) is estimated to be 5 minutes, indicating the average time customer spends on hold.

The standard deviation (σ) is estimated to be 2 minutes, reflecting the variability or dispersion in wait times.

⇒ The area under the P.D.F. curve b/w 0 & 5 minutes gives the exact probability of a customer waiting within that time range.

$$\Rightarrow a=0 \text{ \& \; } b=05$$

$$\therefore P = \int_a^b f(x) dx$$

$$= \int_0^5 f(x) dx.$$

replace the $f(x)$ with the graph function to calculate probability within a range.

